

Ahmed Abdelmotteleb

Experimental Particle Physics Researcher

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EDUCATION

• PhD in Particle Physics (LHCb)

University of Warwick

Oct 2020 – Oct 2024

Coventry, United Kingdom

- Thesis: *Measurement of the W boson properties in the forward region in pp collisions at $\sqrt{s} = 5.02$ TeV.* Supervisors: Prof. Mika Vesterinen, Dr. Miguel Ramos Pernas. [🌐]

• MPhys Physics, Integrated Master's

University of Manchester

Sep 2016 – Jun 2020

Manchester, United Kingdom

- Thesis: *Search for heavy Majorana neutrinos using Left–Right Symmetric models.* Supervisor: Prof. Darren Price.

SELECTED RESEARCH EXPERIENCE

• Postdoctoral Research Fellow (LHCb)

University of Warwick, Department of Physics

Nov 2024 – Present

Coventry, United Kingdom

- Lead analyst for the W boson mass measurement using the full Run 2 dataset at $\sqrt{s} = 13$ TeV; extending my PhD-developed approach to a double-differential measurement in muon transverse momentum and pseudorapidity, with improved momentum-scale and lepton-isolation calibration models.
- Lead analyst for the $B_c^\pm \rightarrow B_s^{(*)0} \pi^\pm$ analysis (Run 1+2) using partial reconstruction; built the machine-learning event selection and the TensorFlow-based **zfit** mass fit.
- Supervised **Warwick URSS** summer projects on conditional normalising flows for muon reconstruction and detector calibration, and quantum-computing applications in HEP (classification, tracking).
- Leading a 20-person analysis team for the TORCH test beams (2022, 2025); overseeing planning, analysis development and code reviews.
- Developed and maintain the **LHCb GitLab Analysis Repository Template**, establishing collaboration-wide standards for reproducible analyses, CI/CD and long-term preservation. It is now the default analysis skeleton across LHCb and part of the official training programme for new collaboration members.
- Physics reviewer for LHCb analyses, including two submitted to Physical Review Letters.

• PhD Researcher (LHCb)

University of Warwick, Department of Physics

Oct 2020 – Oct 2024

Coventry, United Kingdom

- Led the first LHCb measurement of the forward $W \rightarrow \mu\nu$ differential cross-section in bins of muon transverse momentum at $\sqrt{s} = 5.02$ TeV.
- Developed a modular alternative to W boson mass extraction using a reusable, detector-corrected cross-section, enabling future reinterpretation and combinations. The approach is now being adopted by other experiments.
- Co-developed and published a new maximum-likelihood method for template fits, implemented in the widely used **iminuit** Python fitting package.
- Contributed to the experiment's real-time data-processing and trigger software (the system that selects and reconstructs collision events online), through reviewed code merges and performance improvements.
- Presented at the **ICHEP 2024** conference.

PUBLICATIONS AND OUTPUTS

J=JOURNAL, C=CONFERENCE

Co-author to >250 LHCb Collaboration publications ([Inspire HEP profile](#)). Selected outputs with significant personal contributions:

- [J.4] Aaij, R., & others (LHCb Collaboration) (2026). **Measurement of the $W \rightarrow \mu\nu$ cross-sections as a function of the muon transverse momentum in pp collisions at $\sqrt{s} = 5.02$ TeV.** *JHEP*, 2026(03), 148.
- [J.3] Abdelmotteleb, A., & others (2025). **The LHCb Sprucing and Analysis Productions.** *Comput. Softw. Big Sci.*, 9(1), 15.
- [J.2] Aaij, R., & others (LHCb Collaboration) (2024). **Curvature-bias corrections using a pseudomass method.** *JINST*, 19(03), P03010.
- [J.1] Dembinski, H., & Abdelmotteleb, A. (2022). **A new maximum-likelihood method for template fits.** *Eur. Phys. J. C*, 82(11).
- [C.3] Abdelmotteleb, A. (2026). **Electroweak precision measurements at LHCb.** *Progress in High Energy Physics*, 4(1). Proc. PIC 2025.
- [C.2] Abdelmotteleb, A., & others (2026). **Design, assembly and beam validation of a full-scale TORCH module.** *Nucl. Instrum. Meth. A*, 171542.
- [C.1] Abdelmotteleb, A. (2025). **Measurement of single W and Z boson properties in the forward region.** *PoS ICHEP2024*, 360.

SELECTED AWARDS

• Second Place, United Kingdom Quantum Hackathon 2023

National Quantum Computing Centre (NQCC), United Kingdom

Jul 2023



- 2nd place (10 teams) for developing a Quantum Monte Carlo particle transport simulation for nuclear shielding applications.

- Demonstrated expertise in quantum computing, simulation, and interdisciplinary problem solving.

• Stellify Award

University of Manchester, United Kingdom

Jun 2020



- University's most prestigious extracurricular award (<5% award rate).

- Recognised for sustained leadership, community engagement, and academic excellence.

SELECTED LEADERSHIP ROLES

- **QCD, Electroweak and Exotica physics working group convener for Standard Model interactions subgroup** *Jul 2026 – Present*
LHCb, CERN, Switzerland
 - Chair regular QCD, Electroweak and Exotica physics working group meetings on Standard Model measurements, setting agendas and leading analysis discussions.
 - Provide feedback on analysis notes in physics working group review and cover practice talks for upcoming conferences.
 - Support analysts within the subgroup by tracking progress and helping resolve blockers.
- **LHCb Internal Communications Group member** *Jul 2026 – Present*
LHCb, CERN, Switzerland
 - Convene the weekly LHCb General Meeting for >100 participants, coordinating with spokespersons, run coordinators, and internal and external speakers.
 - Improve dissemination of collaboration information across LHCb and collect feedback from collaboration members.
- **Data Processing & Analysis Work Package 4 (DPA WP4) Coordinator, Innovative Analysis Techniques** *Jan 2025 – Present* 
LHCb, CERN, Switzerland
 - Lead the group promoting novel analysis methods (machine learning, quantum computing, and advanced statistics) for LHCb.
 - Run a monthly forum bringing in internal and external experts to introduce emerging techniques to the collaboration.
 - Reviewed high-profile papers within my remit, including ones on particle isolation for Run 3 and a quantum computing application for tracking, the latter of which has been submitted to *Nature*.
 - Initiated the LHCb-CERN QTI (Quantum Technology Initiative) collaboration, enabling IBM Quantum access for LHCb.
 - Organised the first [LHCb Quantum Computing Workshop at Warwick](#) and co-organised the [LHC Quantum Computing workshop](#), connecting high energy physics researchers with CERN QTI and industry partners.
- **LHCb Starterkit Organiser/Teacher** *Sep 2021 – Present* 
LHCb, CERN, Switzerland
 - Developed educational workshop materials and recruited volunteers for 15+ sessions aimed at new collaboration members.
 - Delivered Python and Git/Gitlab workshops to >100 PhD students and postdocs.
- **QCD, Electroweak and Exotica Physics Working Group Liaison** *Feb 2025 – Jul 2026*
LHCb, CERN, Switzerland
 - Facilitated collaboration between physics analysis and real-time analysis/data processing and computing teams; contributed to trigger validation and cross-repository software reviews.

SELECTED TALKS

I=INVITED TALK, C=CONFERENCE, L=LECTURE

- [I.5] **Measuring the W boson mass at LHCb.** University of Lancaster Elementary Particle Physics Seminar (Jan 2026).
- [I.4] **Measuring the W boson mass at LHCb.** University of Warwick Elementary Particle Physics Seminar (Nov 2025). 
- [I.3] **m_W extraction from p_T^ℓ cross-section measurements.** LHC Electroweak Working Group General Meeting (Nov 2025). 
- [I.2] **QCD, Electroweak and Exotica Physics Working Group summary.** LHCb General Meeting (Mar 2025). 
- [I.1] **Introduction to Quantum Computing.** CERN openlab Summer Student Lecture Programme (Jul 2023). 
- [C.3] **Recent electroweak results at LHCb.** Physics in Collision 2025 (Oct 2025). 
- [C.2] **Precision measurements of Z boson production (with a focus on the weak mixing angle).** Standard Model at the LHC (SM@LHC) 2025 (Apr 2025). 
- [C.1] **Measurement of single W and Z boson properties in the forward region.** International Conference on High Energy Physics (ICHEP 2024) (Jul 2024). 
- [L.2] **Introduction to Quantum Computing.** LHCb Quantum Computing Workshop (Jul 2025). 
- [L.1] **Quantum Computing.** Inverted CERN School of Computing (Mar 2023). 

SELECTED SKILLS

- **Machine learning and AI:** PyTorch, TensorFlow, scikit-learn, XGBoost, normalising flows, simulation-based inference.
- **Statistics and analysis:** likelihood fitting, detector calibration, unfolding, uncertainty quantification, systematics modelling.
- **Programming and HEP software:** Python, C++, Julia, Rust, Bash; ROOT, Geant4, Pythia, Sherpa, MadGraph.
- **Computing:** GPU computing (CUDA), Slurm, HTCondor, Kubernetes, Snakemake, CERN LCG Grid.
- **Quantum computing:** Qiskit, PennyLane, variational algorithms, quantum machine learning.
- **Training:** [EDIT Detector School, CERN \(2026\)](#); [Terascale Statistics School, DESY \(2025\)](#); [MCnet School, Durham \(2023\)](#).
- **Languages:** Arabic (native), English (fluent), French (B2), Spanish (B1).

CERTIFICATIONS

- **NVIDIA:** Accelerating CUDA C++ Applications with Multiple GPUs  *Dec 2024*
- **DataCamp:** Machine Learning Scientist with Python  *Jun 2023*
- **NVIDIA:** Fundamentals of Accelerated Computing with Cuda/C++  *Nov 2022*
- **IBM:** IBM Quantum Challenge Advanced Achievement  *Nov 2022*
- **CERN:** CERN School of Computing  *Sep 2022*
- **DataCamp:** Data Scientist with Python  *Sep 2022*
- **Advance HE:** Associate Fellow of the Higher Education Academy  *Aug 2022*
- **University of Warwick:** Academic and Professional Pathway for Postgraduate Students (APP PGR) *Jul 2022*